Which Pass Rate Answers Which Question?

AP Statistics · Unit 2 · Topic 2.2 · TEACHER KEY

CED TETHER

- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.Q** — Calculate statistics for two categorical variables. [Skill 2.C]
- **EK UNC-1.Q.1** — The marginal relative frequencies are the row and column totals in a two-way table divided by the total for the entire table.
- **EK UNC-1.Q.2** — A conditional relative frequency is a relative frequency for a specific part of the contingency table (e.g., cell frequencies in a row divided by the total for that row).
- **LO UNC-1.R** — Compare statistics for two categorical variables. [Skill 2.D]
- **EK UNC-1.R.1** — Summary statistics for two categorical variables can be used to compare distributions and/or determine if variables are associated.

Essential question: How do the denominator and study design limit what a percentage can say?

DOK-3 skill: 4.B · **FRQ pattern:** `marginal-vs-conditional-association-is-not-cause`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Maps the two rates to all students and tutoring users
- **reasoning-2** (required) — Corrects the mixed-denominator comparison with 88 versus 70.7 percent and a 17.3-point group gap
- **reasoning-3** (required) — Explains why self-selection prevents the causal claim

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Treating 75 percent as the pass rate among nonusers
- Subtracting a conditional rate and marginal rate as though they were the two comparison groups
- Calling the 17.3-point association causal because the users performed better

[STAR] THE PROBLEM — annotated

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because* $88\% - 75\% = 13$ *percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE — one sentence before you work the parts

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes. **Key:** Not graded. Expect some students to believe two different pass rates cannot both be correct until their denominators are named.

[BOOK] NAME THE DENOMINATOR, THEN THE CLAIM (keep on the page)

A **marginal relative frequency** uses a row or column total divided by the table total. A **conditional relative frequency** uses a cell count divided by the total for the named group. To decide whether two categorical variables are associated, compare the same response's conditional percentage across the groups. Different conditional distributions show association. They do not show cause when people choose their own groups; a fair causal comparison requires random assignment.

DOK 1 **(a)** Label 75 percent and 88 percent as marginal or conditional. Name the group for each.

Key: The 75% is the marginal relative frequency of passing among all 300 students. The 88% is the conditional relative frequency of passing among students who used the tutoring center; the condition is used the center.

DOK 2 **(b)** Verify the 75 percent and 88 percent rates. The nonuser pass rate is about 70.7 percent; find the user-minus-nonuser gap.

Key: Overall: $225/300 = 75\%$. Users: $66/75 = 88\%$. The group gap is about $88 - 70.7 = 17.3$ percentage points.

DOK 3 **(c)** Can Mara and Noah both be right? Explain their denominators. Use the two group rates to correct Elena's 13-point comparison, and explain why her word raises goes beyond these data. Write 3–4 sentences.

Key: Both rates are correct: Mara uses all 300 students, while Noah uses the 75 tutoring users. Users passed at 88% versus about 70.7% for nonusers, so the observed group gap is about 17.3 percentage points. Elena subtracts the overall rate from the user rate; the overall group includes users and nonusers, so 13 points is not the direct group gap. Students chose tutoring, so differences such as preparation could help explain the association; the table does not show tutoring raised pass rates.